

## Ejercicio 2

Función:  $f(x)=x^2-4$  Intervalo inicial:  $[0,3]$

| Iteración | a      | b      | c      | f(a)    | f(b)   | f(c)    |
|-----------|--------|--------|--------|---------|--------|---------|
| 1.0000    | 0.0000 | 3.0000 | 1.5000 | -4.0000 | 5.0000 | -1.7500 |
| 2.0000    | 1.5000 | 3.0000 | 2.2500 | -1.7500 | 5.0000 | 1.0625  |
| 3.0000    | 1.5000 | 2.2500 | 1.8750 | -1.7500 | 1.0625 | -0.4844 |
| 4.0000    | 1.8750 | 2.2500 | 2.0625 | -0.4844 | 1.0625 | 0.2539  |
| 5.0000    | 1.8750 | 2.0625 | 1.9688 | -0.4844 | 0.2539 | -0.1240 |
| 6.0000    | 1.9688 | 2.0625 | 2.0156 | -0.1240 | 0.2539 | 0.0627  |
| 7.0000    | 1.9688 | 2.0156 | 1.9922 | -0.1240 | 0.0627 | -0.0312 |
| 8.0000    | 1.9922 | 2.0156 | 2.0039 | -0.0312 | 0.0627 | 0.0156  |
| 9.0000    | 1.9922 | 2.0039 | 1.9980 | -0.0312 | 0.0156 | -0.0078 |
| 10.0000   | 1.9980 | 2.0039 | 2.0010 | -0.0078 | 0.0156 | 0.0039  |
| 11.0000   | 1.9980 | 2.0010 | 1.9995 | -0.0078 | 0.0039 | -0.0020 |
| 12.0000   | 1.9995 | 2.0010 | 2.0002 | -0.0020 | 0.0039 | 0.0010  |

Resultado:  $x \approx 2.0000$

Nota: se detiene cuando el error es menor que la tolerancia indicada o cuando ya se aproxima a l

| $f(a)*f(c)$ | error  |
|-------------|--------|
| 7.0000      | 3.0000 |
| -1.8594     | 1.5000 |
| 0.8477      | 0.7500 |
| -0.1230     | 0.3750 |
| 0.0601      | 0.1875 |
| -0.0078     | 0.0938 |
| 0.0039      | 0.0469 |
| -0.0005     | 0.0234 |
| 0.0002      | 0.0117 |
| 0.0000      | 0.0059 |
| 0.0000      | 0.0029 |
| 0.0000      | 0.0015 |

a raíz.